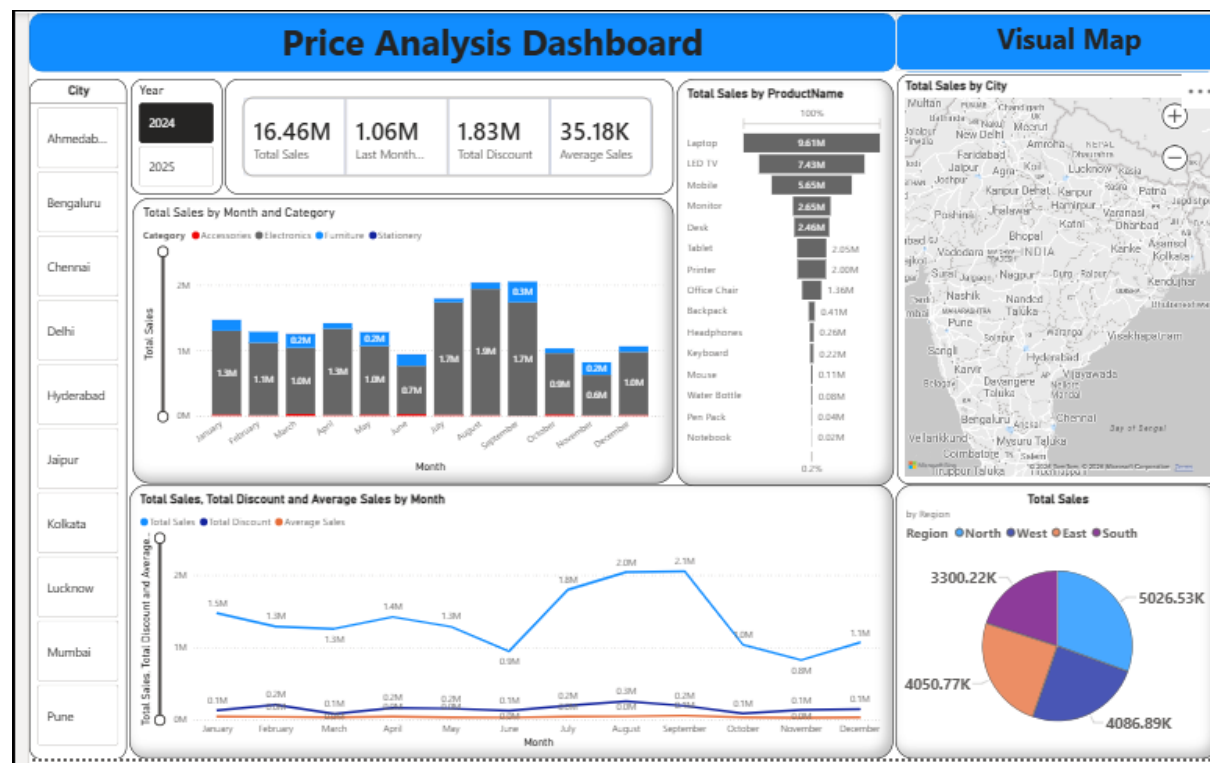


PRICE ANALYSIS DASHBOARD DOCUMENTATION



PRICE ANALYSIS DASHBOARD

End-to-End Power BI Project Documentation

Developed By

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Project Name:

Price Analysis Dashboard

Tool Used

- Power BI Desktop
- Power Query
- DAX
- Data Modeling

1. Project Objective

The objective of this project is to analyze product sales, discounts, customer purchases, employee performance, and regional sales to help management make better business decisions.

The dashboard provides insights into:

- Total Sales

- Total Discount
 - Average Sales
 - Monthly Sales Trend
 - Product Performance
 - Regional Analysis
 - City-wise Sales
 - Payment Analysis
 - Order Status
 - Customer Purchase Behavior
-

2. Business Problem

The company wants to answer the following questions:

- Which products generate maximum revenue?
 - Which cities contribute the highest sales?
 - Which category performs best?
 - How much discount is given?
 - Which region generates maximum sales?
 - Which month has highest sales?
 - What is the average sales value?
 - Which payment method is mostly used?
 - Which employee generated maximum sales?
-

3. Dataset Information

Column	Data Type	Description
OrderDetailID	Integer	Unique order detail ID
OrderID	Text	Order Number
CustomerID	Text	Customer Identifier
CustomerName	Text	Customer Name
EmployeeID	Text	Employee Identifier
EmployeeName	Text	Sales Employee
ProductName	Text	Product Sold
Category	Text	Product Category
Quantity	Integer	Quantity Sold
UnitPrice	Integer	Price Per Unit
GrossAmount	Integer	Sales Before Discount
DiscountPercent	Integer	Discount Percentage
DiscountAmount	Decimal	Discount Given
Sales	Decimal	Final Sales Amount

Column	Data Type	Description
PaymentMethod	Text	Payment Mode
OrderStatus	Text	Delivered / Pending / Cancelled
OrderDate	Date	Order Date
ShipDate	Date	Shipping Date
City	Text	Customer City
Region	Text	Region

4. Data Cleaning (Power Query)

Performed the following transformations:

- Checked Null Values
- Removed Duplicate Records
- Changed Data Types
- Renamed Columns
- Removed Errors
- Standardized Text Format
- Created Date Table
- Checked Data Quality

5. Data Model

Fact Table

Order_Details

Contains transactional sales data.

Dimension Table

DataTable

Relationship:

DataTable[Date]

|
|

Order_Details[OrderDate]

Relationship Type

One-to-Many

Cross Filter

Single Direction

6. Date Table

Created using CALENDAR function.

DataTable =

CALENDAR(

MIN(Order_Details[OrderDate]),
MAX(Order_Details[OrderDate])
)

Additional Columns

Year = YEAR(DateTable[Date])

Month = FORMAT(DateTable[Date],"MMMM")

Month No = MONTH(DateTable[Date])

Quarter = "Q"&FORMAT(DateTable[Date],"Q")

7. KPI Measures

Total Sales

Total Sales =

SUM(Order_Details[Sales])

Total Orders

Total Orders =

DISTINCTCOUNT(Order_Details[OrderID])

Total Quantity

Total Quantity =

SUM(Order_Details[Quantity])

Total Discount

Total Discount =

SUM(Order_Details[DiscountAmount])

Average Sales

Average Sales =

AVERAGE(Order_Details[Sales])

Gross Sales

Gross Sales =

SUM(Order_Details[GrossAmount])

Total Customers

Total Customers =

DISTINCTCOUNT(Order_Details[CustomerID])

Total Employees

Total Employees =
DISTINCTCOUNT(Order_Details[EmployeeID])

Average Discount %

Average Discount =
AVERAGE(Order_Details[DiscountPercent])

Sales Last Month

Sales Last Month =
CALCULATE(
[Total Sales],
DATEADD(DateTable[Date],-1,MONTH)
)

Sales Growth %

Sales Growth % =
DIVIDE(
[Total Sales]-[Sales Last Month],
[Sales Last Month]
)

Average Order Value

Average Order Value =
DIVIDE(
[Total Sales],
[Total Orders]
)

Total Delivered Orders

Delivered Orders =
CALCULATE(
COUNT(Order_Details[OrderID]),
Order_Details[OrderStatus]="Delivered"
)

Pending Orders

Pending Orders =
CALCULATE(
COUNT(Order_Details[OrderID]),
Order_Details[OrderStatus]="Pending"

)

Cancelled Orders

Cancelled Orders =

```
CALCULATE(  
COUNT(Order_Details[OrderID]),  
Order_Details[OrderStatus]="Cancelled"  
)
```

8. KPI Cards Used

Your dashboard displays:

KPI	Description
Total Sales	Overall Revenue Generated
Last Month Sales	Previous Month Revenue
Total Discount	Total Discount Given
Average Sales	Average Sales Per Order

You can also add:

- Total Customers
- Total Orders
- Total Quantity Sold
- Average Order Value
- Sales Growth %
- Total Employees
- Delivered Orders
- Pending Orders
- Cancelled Orders

9. Dashboard Visuals

Slicers

- City
 - Year
-

KPI Cards

- Total Sales
 - Last Month Sales
 - Total Discount
 - Average Sales
-

Clustered Column Chart

Title

Total Sales by Month and Category

Purpose

Compare category-wise monthly sales.

Funnel Chart

Title

Top Sales by Product Name

Purpose

Top-performing products.

Map

Title

Total Sales by City

Purpose

City-wise sales distribution.

Line Chart

Title

Sales Trend

Purpose

Monthly sales trend.

Pie Chart

Title

Sales by Region

Purpose

Regional sales contribution.

10. Dashboard Insights

Example Insights:

- Laptop generated the highest sales.
 - North Region contributed the highest revenue.
 - September recorded maximum sales.
 - Electronics category outperformed others.
 - Delhi generated the highest city sales.
 - Discounts increased during festive months.
 - Average order value remained stable.
 - Cash payments were higher than online payments.
-

11. Business KPIs

KPI	Formula
Revenue	Sum of Sales
Profit Margin	Profit / Sales
Average Order Value	Sales / Orders
Customer Count	Distinct Customers
Sales Growth	Current Month vs Last Month
Discount %	Discount / Gross Sales
Delivery Rate	Delivered Orders / Total Orders
Cancellation Rate	Cancelled Orders / Total Orders

12. Dashboard Features

- ✓ Dynamic Slicers
 - ✓ Interactive Charts
 - ✓ KPI Cards
 - ✓ Drill Down
 - ✓ Cross Filtering
 - ✓ Dynamic Measures
 - ✓ Responsive Layout
 - ✓ Date Intelligence
-

13. Business Benefits

- Monitor overall company sales.
 - Identify top-performing products and categories.
 - Compare city and region performance.
 - Track monthly sales trends.
 - Evaluate discounts and pricing strategies.
 - Analyze employee performance.
 - Improve inventory planning.
 - Enhance decision-making with interactive visuals.
-

14. Future Enhancements

- Forecast Sales using Analytics.
- Dynamic Top N Product Selection.
- Customer Segmentation (RFM Analysis).
- Profitability Dashboard.
- Target vs Actual Sales Analysis.
- Inventory Dashboard.
- AI Visuals in Power BI.

- Automated Refresh via Power BI Service.
- Drill-through Pages for Product and Customer Details.

15. Dashboard Summary

This **Price Analysis Dashboard** is a complete business intelligence solution built using **Power BI**, designed to monitor sales performance, discounts, product trends, customer behavior, and regional performance. It integrates transactional data with a dedicated Date table, applies DAX measures for KPI calculations, and presents interactive visualizations that support strategic decision-making. The dashboard enables users to analyze sales trends, compare category performance, identify top-selling products, and evaluate regional contributions through dynamic filtering and drill-down capabilities.

Suggested Additional KPIs for an Interview-Level Dashboard

KPI	Business Value
Total Sales	Overall revenue generated
Gross Sales	Sales before discounts
Net Sales	Revenue after discounts
Total Orders	Number of orders placed
Total Customers	Unique customers served
Total Products Sold	Total quantity sold
Average Order Value (AOV)	Revenue per order
Average Selling Price (ASP)	Sales per unit sold
Sales Growth %	Month-over-month performance
Total Discount	Total discounts offered
Discount %	Discount as a percentage of gross sales
Top Selling Product	Best-performing product
Top Sales City	Highest revenue city
Top Region	Best-performing region
Delivered Order %	Delivery success rate
Cancellation Rate	Percentage of cancelled orders
On-Time Delivery %	Orders shipped on time (using OrderDate and ShipDate)
Top Employee by Sales	Best-performing salesperson
Category Contribution %	Share of each category in total sales
Customer Retention Rate	Repeat customer percentage

Business Insights

1. Overall Sales Performance

- The company generated **16.46M** in total sales, indicating strong business performance during the selected period.
 - The average sales per transaction are approximately **35.18K**, showing a healthy order value.
-

2. Product Performance

- **Laptop** is the highest revenue-generating product with approximately **9.61M** in sales.
- LED TVs and Mobile Phones are the next top-selling products, making them key revenue drivers.
- Products such as **Notebook, Pen Pack, Water Bottle, and Mouse** contribute very little to overall sales, indicating opportunities for promotional campaigns or inventory review.

Business Recommendation:

- Maintain sufficient inventory for high-demand products.
 - Bundle low-selling products with top-selling products to increase sales.
-

3. Category Performance

- The **Electronics** category contributes the highest percentage of total sales.
- Accessories and Stationery generate comparatively lower revenue.

Business Recommendation:

- Expand the Electronics product range.
 - Review pricing and marketing strategies for low-performing categories.
-

4. Monthly Sales Trend

- Sales remained relatively stable during the first half of the year.
- A significant increase is observed between **July and September**, with **September** recording the highest sales.
- Sales decline noticeably in **October and November** before recovering slightly in December.

Business Recommendation:

- Investigate the factors behind the sales peak (seasonal demand, promotions, festivals).
 - Plan marketing campaigns before the low-sales months to sustain revenue.
-

5. Regional Analysis

- The **North** region contributes the highest share of total sales.
- West and East regions perform moderately.
- The **South** region has the lowest contribution.

Business Recommendation:

- Strengthen sales and marketing activities in the South region.
 - Analyze customer preferences by region to improve product offerings.
-

6. City-wise Performance

- Major metropolitan cities generate higher revenue compared to smaller cities.
- Sales distribution indicates strong demand in urban markets.

Business Recommendation:

- Increase inventory and promotional efforts in top-performing cities.
 - Explore opportunities to expand into underperforming cities.
-

7. Discount Analysis

- The company provided approximately **1.83M** in discounts.
- Discounts represent a significant investment in driving sales.

Business Recommendation:

- Evaluate whether discounts are effectively increasing revenue and customer retention.
 - Apply targeted discounts instead of blanket discount campaigns.
-

8. Customer Purchasing Behavior

- Customers show higher demand for premium electronic products.
- High-value products contribute significantly more revenue than low-cost products.

Business Recommendation:

- Introduce loyalty programs for repeat customers purchasing premium products.
 - Recommend complementary products during checkout.
-

9. Inventory Insights

- Sales are concentrated among a few products.
- Several products have very low sales volume.

Business Recommendation:

- Maintain higher inventory levels for fast-moving products.
 - Reduce stock levels for slow-moving products to lower inventory costs.
-

10. Operational Performance

- Monitoring Order Status enables the business to identify delayed or pending shipments.
- Tracking shipping performance helps improve customer satisfaction.

Business Recommendation:

- Reduce delivery delays.
 - Monitor cancellation and return rates regularly.
-

11. Employee Performance (If Employee Analysis is Added)

- Sales performance can be analyzed by employee to identify top performers.
- High-performing employees can be rewarded, while others can receive additional training.

12. Payment Method Analysis (If Visual Added)

- The most frequently used payment method can be identified.
- Understanding customer payment preferences helps optimize payment options.

Key Business Recommendations

Area	Recommendation
Product Sales	Increase inventory for top-selling products like Laptops and LED TVs.
Low-Selling Products	Introduce promotional offers or product bundles.
Regional Sales	Focus marketing efforts on the South region to improve sales.
Monthly Sales	Launch promotional campaigns before October and November to offset seasonal declines.
Discount Strategy	Optimize discounts to maximize profitability while maintaining sales growth.
Customer Retention	Implement loyalty and rewards programs for repeat customers.
Employee Performance	Track employee sales and recognize top performers.
Inventory Management	Stock fast-moving products adequately and reduce excess inventory for slow-moving items.
Expansion	Explore new markets in underperforming cities and regions.

Executive Summary

The dashboard reveals that the business generated **16.46M** in total sales, with **Laptops** emerging as the top-selling product and the **North** region contributing the highest share of revenue. Sales peaked during **September**, suggesting strong seasonal demand, while **October and November** experienced a decline. Electronics dominate overall sales, whereas Accessories and Stationery contribute less. The company invested **1.83M** in discounts, emphasizing the importance of evaluating promotional effectiveness. These insights enable management to optimize inventory, refine pricing strategies, target underperforming regions, improve operational efficiency, and drive sustainable revenue growth.